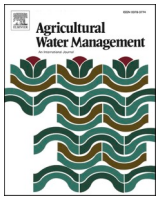
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Towards efficient irrigation management at field scale using new technologies: A systematic literature review

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ABSTRACT

Life on earth is linked to water resources. Recently, alarm bells have been ringing in global organizations to raise awareness of the importance of rational use of water resources, which are becoming an increasingly scarce commodity. The majority of the world's freshwater is used for agricultural irrigation, hence there is a need to adopt an intelligent irrigation strategy that will lead to sustainable agricultural management. To reap the full benefits, irrigation strategy must be accompanied by a good understanding of field characteristics. Several studies have benefited from the improvement of new technologies for irrigation scheduling, but taking only soil water properties as a basis for research, and to our knowledge there is no systematic literature review study to date that aims at irrigation scheduling taking into consideration the characteristics of the crop field for intelligent and efficient agricultural management. This literature review article aims to explore the new Internet of Things and Artificial Intelligence technologies used on the one hand for monitoring and predicting the coefficients that control the crop evapotranspiration process responsible for crop water losses, namely the reference crop evapotranspiration coefficient ETo and the crop coefficient Kc, and on the other hand for a good and intelligent understanding of the: physical, chemical, biological and hydrological characteristics of a specific field, and which affect the crop evapotranspiration process and therefore yield. Following a systematic literature review methodology led us to a refined selection of 55 journal articles for further analysis. We have identified that the profitability of a crop field is closely linked to the right strategies adopted in a specific crop plot, and these strategies can only be defined after a good understanding of the field's characteristics. We were able to discuss these field characteristics through the primary studies which enabled us to develop an intelligent model that brings together the different approaches adopted for irrigation scheduling and farm management and to identify gaps and limitations in the use of new technologies for farm management at field scale, and thus pave the way for further research.

1. Introduction

According to the World Bank report, food security is guaranteed by the agricultural sector, which constitutes one of the economic pillars in the majority of the world's countries (Burki, 2022). Intensifying production and increasing agricultural yields are closely linked to good irrigation practices, given the direct positive effect of water on crop income (DeJonge et al., 2007). Excessive use of water resources, especially in the agricultural sector, which is responsible for consuming most of the world's freshwater, is making these resources less and less available. In addition, the effects of climate change have aggravated the situation, making rational use of water resources essential (DeJonge et al., 2007; Burki, 2022; Kocian et al., 2023). Adopting a sustainable agricultural management strategy that achieves the highest yields without depleting natural resources is a solution strongly encouraged by the Food and Agriculture Organization of the United Nations, and one in which all countries of the world must be involved (Burki, 2022).

Closed-loop irrigation (Bwambale et al., 2022), also known as smart irrigation (Obaideen et al., 2022), is a promising solution that harnesses new Internet of Things and Artificial Intelligence technologies to manage and schedule irrigation, applying water at the optimum time according to crop needs. Crop water requirements are influenced by the

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process of crop evapotranspiration, which includes two inseparable phenomena responsible for water losses: soil evaporation and crop transpiration (Anderson and French, 2019).

The indirect dual crop coefficient method suggested by (Anderson and French, 2019) is the standard method used for precise irrigation scheduling. This method is essentially centered on the calculation of the crop evapotranspiration coefficient ETc, which enables the estimation of the exact amount of water to be applied by taking into consideration the meteorological conditions summarized in the reference crop evapotranspiration coefficient ETo and the phenological conditions of the plants estimated by the crop coefficient Kc (Abou Ali et al., 2023; Feng et al., 2023; Gonçalves et al., 2023; X. Han et al., 2021; Vories et al., 2021). Several new studies have benefited from the development of new technologies for estimating the evapotranspiration coefficient of ETc crops. For example, (Zhangzhong et al., 2023) proposed a mobile solution for real-time monitoring of the irrigation requirements of certain types of lettuce grown in greenhouses using sensors and machine learning algorithms. (Kocian et al., 2023) have suggested an Internet of Things-based solution for monitoring water use for crops grown in greenhouses. However, estimating the water requirements of crops grown in controlled environments is much simpler than estimating them in arable fields, since several factors enter into the formation and evolution of the evapotranspiration process. (Wu et al., 2023) attempted to estimate, in a semi-humid region in China, the irrigation requirements of maize by training meteorological data from a local weather station and crop data measured by a lysimeter, in machine learning models, while (Y. Zhang et al., 2023) used crop data acquired by sensors based on an unmanned aerial vehicle (UAV) to estimate the daily ETc of maize.

Using the dual crop coefficient method to estimate irrigation requirements can be an effective but not efficient solution if it is not accompanied by a prior and accurate understanding of the climatic, physical, chemical, biological and hydrological characteristics of the arable field. By way of example, if farmers grow a type of crop in a field that is not compatible with the field's characteristics, the best possible irrigation strategy in this case is not necessarily the efficient solution. On the other hand, the solution that consists in changing the type of crop, according to the field's characteristics, and accompanying it with good irrigation management is necessarily in this case the efficient solution in terms of managing natural resources and obtaining the best possible yield in the field under study. (Karimzadeh Soureshjani et al., 2019) studied the existing relationship between crop yield and air temperature and came to the result that management practices such as adjusting planting date can be applied by knowing the critical periods during which crops are most vulnerable to climate change conditions.

To the best of our knowledge, to date there has been no literature review study that, in addition to bringing together the key factors affecting the evapotranspiration process in a field, explores the new Internet of Things and artificial intelligence technologies used to monitor, detect, locate and predict these factors, which are the climatic, physical, chemical, biological and hydrological characteristics of a field with a view to efficient irrigation management.

2. Related work

Intelligent and sustainable irrigation management has become a necessity whether in smart cities or agricultural fields. (Gimpel et al., 2021) proposed an irrigation system for urban trees using Internet of Things (IoT) architecture technologies and meteorological data. The proposed solution achieved promising results in terms of reducing irrigation water consumption.

(Veerachamy et al., 2022) have proposed an irrigation monitoring and control system using IoT technology. The suggested solution can be managed manually through the analysis of data displayed on a mobile app, or automatically by determining the threshold soil moisture value of the field to be irrigated. The system can be enhanced by the introduction of Artificial Intelligence algorithms as used by (R et al., 2023) who proposed an autonomous system based on Artificial Intelligence and IoT-6 G that both predicts climate change and precipitation patterns and measures soil moisture levels for the coming days. The system has been tested in a controlled environment, achieving high accuracy in soil moisture prediction.

(Nawandar and Satpute, 2019) have proposed a system that takes into consideration crop data such as: soil type, crop type as well as its planting date, and some field data to calculate evapotranspiration and make irrigation decisions based on neural methods. (Xie et al., 2022) proposed a system for irrigation control and soil moisture level monitoring in a Litchi orchard. The system used data from a soil moisture sensor and data from a weather station as inputs to an improved Particle swarm optimization (PSO) algorithm. The results showed that the system is highly accurate, easy to implement and applicable in real time. (Alibabaei et al., 2022) studied the ability of reinforcement learning to schedule the irrigation of a tomato field without affecting productivity. They proposed a deep Q-network model (LSTM) that indicates when and how much to irrigate. The proposed model increased productivity and reduced water consumption.

(Cordeiro et al., 2022) proposed deep neural network architectures using Long short-term memory (LSTM) and Bidirectional LSTM (BiLSTM) to develop economical soil moisture prediction models. The problem of missing input data was also addressed by using the k-nearest neighbors (KNN) algorithm to fill in the unknown values and guarantee a high degree of reliability. One of the perspectives of this work is the evaluation of the energy consumption of fog nodes. (Maria Manuel Vianny et al., 2022) proposed an irrigation system for a banana field that reduces energy and water consumption through the use of IoT sensors, a storage system and the KNN, Gradient boosting and LSTM algorithms. (Goap et al., 2018) have also proposed a water and energy-efficient system that relies on field sensor data and weather forecast data as inputs to a method developed based on the Support Vector Regression (SVR) algorithm and K-means clustering. One of the perspectives of this work is to further develop the system so that it works with multiple sensor nodes while minimizing system cost. (Khan et al., 2022) have proposed a smart irrigation solution that is energy-efficient, reliable and secure. They have used methods based on Artificial Intelligence to select the optimal location of sensor nodes, to regulate energy and to choose the most optimal routing path. (Al-Ali et al., 2019) have designed a prototype solar energy system based on IoT and a fuzzy logic algorithm for irrigation control and decision-making. Table 1 shows more details of the irrigation systems studied.

Another technology that can be beneficial in irrigation management and scheduling is the Digital Twin. (Alves et al., 2023) used this technology to develop a closed-loop monitoring and control system for irrigation. The proposed solution was used to evaluate the behavior of an IoT platform and irrigation strategies before applying them on the agricultural field. The results show that the system is capable of collecting data to create a daily irrigation schedule.

3. Research methodology

A targeted, relevant literature review leads to strong and solid results. The methodology of systematic literature review SLR is widely used to conduct a successful literature search.It is a method of determining, assessing and evaluating all available research relevant to a particular research question or phenomenon of interest (Kitchenham et al., 2009). The SLR guidelines used in this document are those of (Kitchenham et al., 2009), which have been adapted to software engineering from the medical literature. The SLR protocol defined in this article was adapted from the protocol used by (Nyoman Kutha Krisnawijaya et al., 2022; Tummers et al., 2019), also based on (Kitchenham et al., 2009). The identification of the research questions was our first step, followed by the search protocol definition which include determining the search string and the sources to be consulted. The definition of the search protocol was an iterative process, in which the results of

Table 1
Related work using new technologies in irrigation management.

Reference	Goal	Results	Future works
(Veerachamy et al., 2022)	Proposal for an automated irrigation control system using field sensors and a Cloud platform.	The system saves water and labor in the irrigation process.	- Due to poor network coverage, the system can malfunction, making it necessary to improve it. - The system can be improved by introducing machine learning algorithms and other input parameters for more reliable forecasting.
(R et al., 2023)	Development of an efficient crop irrigation methodology using artificial intelligence and IoT-6 G technologies.	The proposed system was tested in a controlled environment and achieved an accuracy of 86.34%.	The solution needs to be tested in a real environment with several field characteristics.
(Nawandar and Satpute, 2019)	Development of a low-cost intelligent irrigation system that enables irrigation scheduling and decision-making.	The system was tested on spinach seeds during their growing period, and achieved water savings of around 67% compared with traditional methods.	Future work will focus on the application of such a system to other types of crop and in other environments.
(Xie et al., 2022)	Design of an irrigation control system using soil moisture level monitoring in a Litchi orchard.	The average soil moisture value estimated by the developed method is closer to the optimum soil moisture defined by the system.	- The generalizability of the system can be improved by carrying out this study on different types of regions or orchards. - The spatial distribution of soil moisture needs to be taken into account for more accurate measurements.
(Alibabaei et al., 2022)	Study of the ability of reinforcement learning to schedule the irrigation of a tomato field without affecting productivity.	Compared with fixed-base and threshold-based irrigation methods, the trained model increased productivity by 11% and reduced water consumption by 20–30%.	The input data for the model developed were simulated, hence the need to train the model with field data to make it more realistic.
(Cordeiro et al., 2022)	Construction of soil moisture prediction models for coconuts and cashews based on deep neural network architectures.	The real experiment showed promising results, with more accurate predictions in the deepest soil layers than in the surface layers.	Evaluation of the energy consumption of fog nodes.
(Maria Manuel Vianny et al., 2022)	Optimizing water and energy use in banana agriculture using IoT and machine learning algorithms.	31.14% of water consumption is reduced in a period characterized by low rainfall.	External interference such as signal noise must be eliminated to improve prediction accuracy and decision-making.
(Goap et al., 2018)	Development of a soil moisture prediction algorithm using field sensor data and weather forecast data.	A web interface has been developed to view sensor data in real time and plan irrigation and the results of the prediction system are very encouraging.	Further develop the system to work with multiple sensor nodes while minimizing system cost.
(Al-Ali et al., 2019)	Irrigation management using a renewable energy source to provide the energy needed to keep the system running smoothly.	The prototype provided excellent irrigation using only solar energy.	Implement such a system in a real agricultural environment.
(Alves et al., 2023)	Creation of a simulation model to simulate and evaluate the behavior of an irrigation system defined in an application scenario.	The results show that the system can increase water use efficiency by evaluating several irrigation strategies with data collected from multiple sources.	The study doesn't use actual data from a farm, so in the future, it's important to evaluate the behavior of the IoT platform in a real environment in order to validate the communication between the devices and the platform.

one search string were used to adjust the next search string. Defining the criteria for selecting the studies was our third step. In the fourth step, we applied the criteria to the search results. After this step, we obtained the articles we used for an in-depth evaluation based on a quality assessment method. The fifth step involved extracting useful data from the selected articles. The third, fourth and fifth steps were iterative. Finally, in the sixth stage, we defined the method used to synthesize and present the data.

3.1. Research questions

Irrigation compensates for crop evapotranspiration losses, so several research questions arise from this starting point:

- RQ1: What new technologies are being used to estimate and predict the ETo climate parameter?
- RQ1.1: Does a good prediction of this coefficient guarantee good irrigation scheduling? In other words, is irrigation intensification always the optimal solution to compensate for the effect of soil evaporation, or do we need to think differently?
- RQ2: What factors have a significant influence on the crop coefficient Kc as mentionned in FAO Irrigation and Drainage Paper?
- RQ2.1: What effect do soil physical properties have on agricultural management in general? What new technologies are being used in this field of research?
- RQ2.2: What effect do soil chemical properties have on agricultural management in general? What new technologies are being used in this field of research?
- RQ2.3: What are the soil's hydrological properties to be taken into account when managing irrigation
- RQ2.3.1: What new technologies are being used to estimate soil moisture
- RQ2.3.2: What is the effect of groundwater on crop growth, and how can we exploit new technologies to estimate and predict groundwater levels

- RQ2.3.3: What is the effect of hydraulic conductivity on irrigation scheduling, and how can we exploit new technologies to estimate and predict hydraulic conductivity?
- RQ3: What is the effect of non-standard elements such as weeds and plant diseases on crop development and yield
- RQ3.1: What new technologies are being used to manage weeds
- RQ3.2: What new technologies are being used to manage plant diseases
- RQ4: Do the farming practices adopted in a field affect the crop coefficient, and if so, how does this manifest itself

The first research question is addressed by identifying new technologies used for short, medium and long-term ETo estimation and prediction. This question has generated another sub-question which highlights the solution of irrigation intensification to compensate for the effect of climatic parameters on the soil and discusses the feasibility of this solution in cases of climate changes. This sub-question is addressed by analyzing case studies that investigate the effect of climate changes such as temperature on irrigated crop yields. The answer to this question led us to the second research question, which discusses the soil properties that influence crop evapotranspiration, namely soil physical, chemical and hydrological properties. This question is followed by three sub-questions. The first sub-question is addressed by identifying new technologies used for the prediction and classification of soil textures and their irrigation requirements. The second sub-question is addressed by identifying new technologies used for the prediction and real-time monitoring of soil chemical properties. The third sub-question, which discusses soil hydraulic properties, has been addressed firstly by identifying new technologies used for the short and long-term prediction of soil moisture, secondly, by identifying new technologies used for the short and long-term prediction of groundwater levels, and thirdly, by identifying new technologies used for the prediction and mapping of saturated and unsaturated hydraulic conductivity. The third research question, which discusses the effect of non-standard elements such as pests and weeds on crop evaporation, was addressed firstly by identifying new technologies used for weed detection, localization and prediction, and secondly by identifying new technologies used for plant disease detection and severity. Finally, the fourth research question, which discusses the effect of agricultural practices on crop evapotranspiration, was developed by analysing some case studies that investigate the effect of sometimes inappropriate agricultural practices such as the mode of application and the quantity of water on crop yield.

3.2. Search protocol

To answer the research questions, we carried out a systematic search of the literature available in digital libraries that publish high-quality articles such as IEEE Xplore, Science Direct, Springer, Web of Science and Scopus. The use of new technologies in agriculture has become increasingly frequent in recent years. Consequently, literature from the last eight years (2015–2023) was used in this study. The targeted articles were journal articles. In order to find as many relevant results as possible, an automated search and a manual search were carried out. The automated search was carried out using search strings in the digital libraries listed above. The syntax of each database is different. An example of the general syntax of the search string is illustrated by the following query: ("Smart irrigation " OR " Irrigation planning ") AND ("Reference crop evapotranspiration " OR " Soil classification " OR "Soil chemical properties") AND ("Artificial intelligence " OR " Machine learning " OR " Internet of things "). A manual search was carried out by manually going through the list of references of articles discovered by the automated search (snowball effect). Table 2 shows the keywords used to find relevant articles

The search results are listed in the second column of Table 3. The source containing the highest number of studies was Science Direct with 413 studies, and the source containing the lowest number of studies was Springer with 19 studies. In total, 625 articles were identified by the automated search and 47 articles were added by the manual search.

Table 2
List of keywords used by category.

Study problem	Factors	New Technologies
"Smart irrigation'' OR "Irrigation planning''	" Crop evapotranspiration'' OR " Reference crop evapotranspiration'' OR " Soil classification'' OR "Soil texture'' OR "Soil chemical properties'' OR "Soil moisture'' OR "Groundwater level'' OR "Goundwater quality'' OR "Saturated hydraulic conductivity'' OR "Unsaturated hydraulic conductivity'' OR	"Artificial intelligence'' OR "Machine learning'' OR "Deep learning'' OR "Internet of things''
	"Weed detection'' OR "Weed classification'' OR	OR
	"Disease detection'' OR "Disease classification'' OR "Agricultural practices'' OR "Crop coefficient''	IoT

Table 3
Overview of search results and study selection.

Sources	After Automated and manual search	After applying selection criteria
Science Direct	413	43
Springer	19	0
IEEE Xplore	60	2
Scopus	108	11
Web of Science	25	0
Manual search	47	14
Total	672	55

3.3. Selection criteria

Journal articles, conference papers and book chapters were potentially provided after applying the search strings. Based on the search results, we applied the selection criteria presented in Table 4 to identify the most relevant studies. First, the selection criteria were applied by choosing only journal articles and reading the title, keywords and abstract of the studies. Next, we proceeded to a detailed reading of the studies and the application of exclusion criteria. After applying these criteria, 55 of the 672 articles were retained. The selection process is shown in Table 3.

3.4. Quality assessment

We also assessed the quality of the 55 studies before moving on to the data synthesis stage. In this research, quality assessment was part of data extraction. When reading the articles, the quality criteria presented in Table 5 were applied as adopted by (Kitchenham et al., 2009) and other SLR researches. Studies were ranked according to the quality of their

Table 4
List of selection criteria.

No	Criteria
EC 1	Articles for which full text is not available
EC 2	Non-English papers
EC 3	The double publication found in several sources
EC 4	Papers do not address research questions.
EC 5	Papers whose proposed study has not been validated

Table 5
List of quality criteria.

No	Question	Yes (1)	Partial (0.5)	No (0)
Q1	Are the study's objectives clearly defined?			
Q2	Is the study's scope and context clearly stated?			
Q3	Are the study variables valid and reliable?			
Q4	Is the research process properly documented?			
Q5	Are all the questions in the study answered?			
Q6	Are negative results reported?			
Q7	Are the main conclusions clearly presented?			
Q8	Are the conclusions coherent with the study's objective?			

final product, and a three-point scale (yes, partial, no) was used for each criterion found during the process. At this stage, articles scoring less than four points out of eight were excluded. Fig. 1 shows that all primary studies obtained a score between five and seven after evaluation of all articles. Consequently, no exclusion decision was made regarding these studies, and all the 55 studies will be used in the data extraction process.

3.5. Data extraction

To answer the research questions, it is necessary to read the 55 primary studies carefully and in full. For this reason, a data extraction table was developed to collect and retrieve all relevant information from these studies. The development of the data extraction table followed an iterative process. First, we rigorously read several arbitrarily selected articles to create the data extraction table. Then, the initial data extraction table was used on other selected articles to extract the data, and if it turned out that the table still didn't cover all the information, we updated it and used the revised table on subsequent articles. This process was repeated. Finally, our final data extraction table includes general information such as study title, authors, year of publication and place of publication. It also contains specific information that directly answers the research questions such as the target domain, the main objectives of the study, the material used, the artificial intelligence algorithms used, the results and the limitations of the study. Details of the data extraction table are provided in the Table 6 in appendix A.(Table 6)

3.6. Data synthesis

Data synthesis is the process of gathering and summarizing the extracted data in a way that answers the questions posed. To achieve this, we tried to extract the obstacles and limitations of each study and to search, in the selected primary articles, for other solutions and innovations capable of filling the gaps found. We used tabular representation of the data where possible, which enabled us to make comparisons between studies. In this way, we were able to extract the future paths of research that remain to be explored.

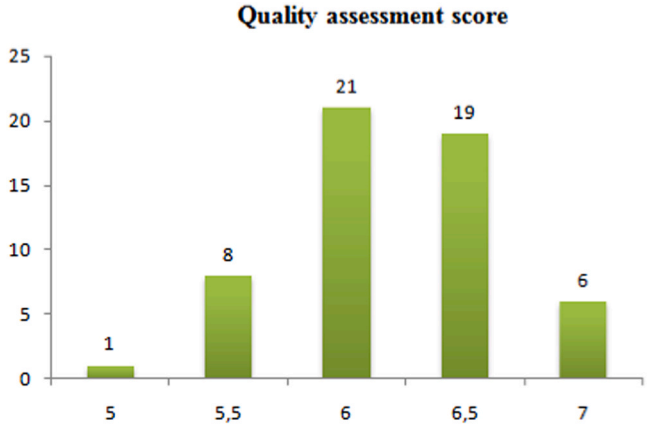


Fig. 1. Distribution of quality scores for selected articles.

Table 6
The data extraction table used in this SLR.

	Extraction element	Contents
General information		
1	Title	Full article title
2	Year	Year of publication
3	Authors	
4	Repository	Science Direct, Springer, IEEE Xplore, Scopus,Web of Science
Study description		
5	Targeted domain	
6	Study motivation	
7	Main aims of the study	
8	Study area	
9	Data used	
10	Range of data	
11	Material used to acquire data	
12	Data processing methods	
13	Artificial intelligence algorithms used	
14	Findings	
15	Limitations	
16	Future Works	
Evaluation		
17	Personal note	Reviewer opinion about the study
18	Additional note	Publication details
19	Quality assessment	Quality scores

4. Results

This section presents the results of data extraction and synthesis. The first sub-section presents the general statistics of the primary studies, and the following sub-sections discuss the results corresponding to the research questions.

4.1. General statistics

The 55 primary studies included in this search are listed Table 7.

Fig. 2 shows the distribution by year of the studies selected between 2015 and 2023 with the majority corresponding to articles from the last four years, as the use of new technologies in agriculture has become increasingly common in recent years.

Fig. 3 shows that China, Iran and the USA are the main contributors having produced the greatest number of documents, with 20, 7 and 5 respectively. They are followed by Brazil, Australia, Japan, the Netherlands, India, Italy and other countries not mentioned here.

Fig. 4 shows the top five journals from which we extracted the articles to be studied. Journal of Hydrology, Computers and Electronics in Agriculture and Agricultural Water Management are the most important, with 9, 9 and 8 respectively, followed by Soil and Tillage Research and Geoderma with 3 each.

4.2. Research questions analysis

The ultimate goal of our study is intelligent irrigation management. We have tried through the research questions to build up a holistic idea of almost all the characteristics of a field and how they affect and interact with plantations to ensure efficient irrigation planning.

Our research questions are progressive, the answer to one question generates another. In this section, we will attempt to answer these questions by analyzing and criticizing previous works.

Table 7
The 55 primary studies used as input for the SLR.

(Di Nunno and Granata, 2023)	2023	(A et al., 2022)	2022	(H. Kashani et al., 2020)	2020
(Gong et al., 2021)	2021	(Abowarda et al., 2021)	2021	(Aguilera et al., 2022)	2022
(Lu et al., 2023)	2023	(Chandrappa et al., 2023)	2023	(Rezaei et al., 2023)	2023
(Maroufpoor et al., 2020)	2020	(Masrur Ahmed et al., 2021)	2021	(W. Zhao et al., 2023)	2023
(Sharafi et al., 2023)	2023	(Datta and Faroughi, 2022)	2022	(Gao et al., 2021)	2021
(Wu et al., 2022)	2022	(Togneri et al., 2022)	2022	(Dang et al., 2023)	2023
(L. Zhao et al., 2022)	2022	(Bennett, 2023)	2023	(Luo et al., 2023)	2023
(Y. Zhang et al., 2020)	2020	(Cai et al., 2022)	2022	(Ruigrok et al., 2023)	2023
(Becker et al., 2023)	2023	(Cai et al., 2021)	2021	(Janarthan et al., 2022)	2022
(Karimzadeh Soureshjani et al., 2019)	2019	(Chen et al., 2022)	2022	(Pal and Kumar, 2023)	2023
(Al-Naji et al., 2021)	2021	(Gharehbaghi et al., 2022)	2022	(Guo et al., 2022)	2022
(Inazumi et al., 2020)	2020	(M. Han et al., 2015)	2015	(Yang et al., 2023)	2023
(J. and T., 2023)	2023	(Liu et al., 2022)	2022	(Souza et al., 2020)	2020
(Kurtulmuş et al., 2022)	2022	(Sheikh Khozani et al., 2022)	2022	(Sawadogo et al., 2023)	2023
(Andrade et al., 2020)	2020	(Veloso et al., 2022)	2022	(Shao et al., 2021)	2021
(Hatanaka et al., 2022)	2022	(Tian et al., 2018)	2018	(Shao et al., 2023)	2023
(Lin et al., 2022)	2022	(D'Emilio et al., 2018)	2018	(W. Zhang et al., 2022)	2022
(Srisomkiew et al., 2021)	2021	(Mozaffari et al., 2022)	2022		
(Xiao et al., 2023)	2023	(Albalasmeh et al., 2022)	2022		

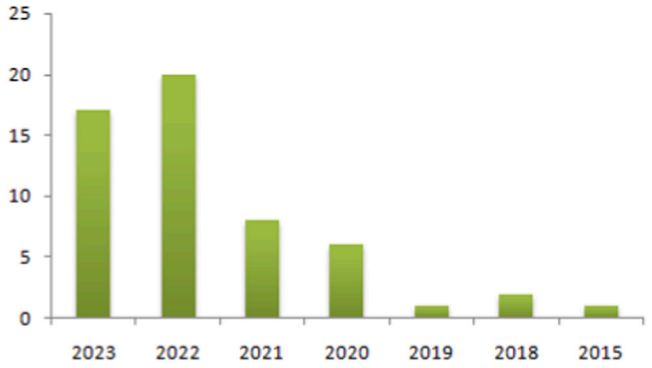


Fig. 2. Year of publication of the primary studies.

4.2.1. RQ-1: What new technologies are being used to estimate and predict the ETo climate parameter?

The total amount of water lost through evaporation per unit area of soil where crops are grown is known as reference crop evapotranspiration (Hunasigi et al., 2023). ETo is a complex and non-linear parameter that is therefore difficult to measure and is influenced by several meteorological factors, which, combined together, affect the soil evaporation process.

ETo is generally evaluated by means of empirical and semi-empirical equations in which temperature, wind speed and humidity are forming some of the key climatological data that provide the energy required for evaporation and remove water vapour from the evaporating surface (Anderson and French, 2019). Fig. 5 shows some of the meteorological

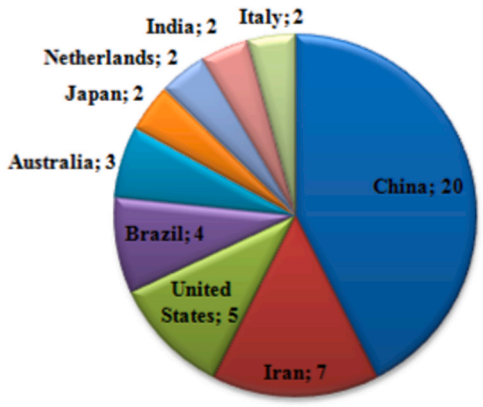


Fig. 3. Number of studies carried out by the main contributing countries.

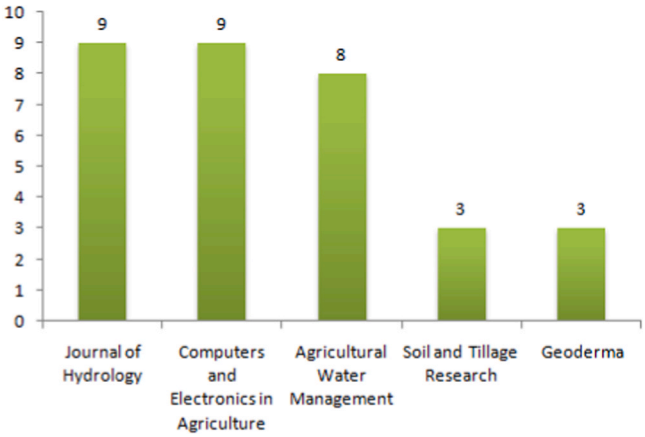


Fig. 4. Top five journals used in research.

factors that influence the value of the ETo parameter.

The development of new technologies has greatly helped researchers to move from assessment to estimation and prediction of this coefficient by combining empirical equations in artificial intelligence algorithms to make short and long-term predictions of ETo in order to fully understand this phenomenon in the field and therefore manage the resources needed to make up for soil evaporation losses.

(Lu et al., 2023) proposed a hybrid prediction model based on signal processing methods combined with the propagation neural network (BPNN) model to increase the reliability of ETo time series estimation. The study achieved very good estimation results for a period of up to seven days, but after that, the model's performance began to degrade. To improve the study, an analysis of the contribution rate of various meteorological factors can be included, as used by (L. Zhao et al., 2022) who relied on the KNN algorithm to extract the contribution rate of meteorological factors and use those with a high contribution rate as inputs to the Elman forecast model which has been optimized by optimization algorithms. The proposed model presented high accuracies only in regions where temperature factors change relatively little throughout the year, a feature that does not affect model calculations. We can conclude that another algorithm other than Elman should be chosen for other regions with considerable variations in input factors in order to improve ETo's daily prediction study.

Another method of determining the meteorological factors that have a major influence on a study area is to make a comparison between empirical equations for calculating ETo as used by (Wu et al., 2022) who, after dividing China into four climate classes, made a comparison between twelve empirical equations to find the most accurate equation for use with machine learning algorithms to predict ETo.The results

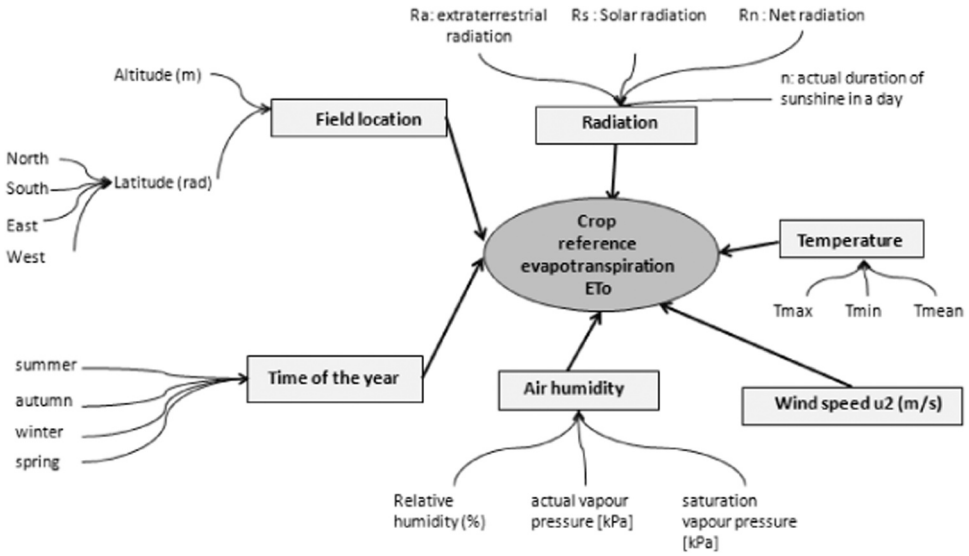


Fig. 5. Meteorological factors influencing the ETo parameter.

showed that the Penman model presented the best estimate of ETo across China.(Sharafi et al., 2023) made a comparison between three empirical equations; BARO, JEHA and Penman in Iran. The results showed that machine learning models based on the combination of input variables from the three equations obtained the best accuracies, especially in semi-arid climates. One recommendation suggested by the authors consists in using relative humidity as a crucial parameter in ETo prediction for humid regions.

The aim of the articles discussed above was to estimate ETo on a daily basis. (Di Nunno and Granata, 2023) took a different approach to predict long-term ETo in areas of Italy. They relied on historical climate parameters (Tmax and Tmin) taken from historical CORDEX data, and future ones based on two future scenarios of greenhouse gas concentration trajectories combined with machine learning algorithms for long-term ETo prediction taking into account the impact of climate change.

Since the study was based solely on historical and future temperatures and extraterrestrial radiation Ra, it can be very useful in identifying the most sensitive areas in terms of potential water deficit in the future rather than estimating ETo. In addition, the use of data from weather stations near the study area could have resulted in more reliable climatic variables. Table 8 gives more details of the systems studied.

We can conclude from all these articles that, to date, there is no universal model for estimating ETo and therefore the effect of soil evaporation in different areas having the same climate. Models need to be developed for each region of the world, indicating the best input

Table 8
Studies using new technologies to estimate and predict ETo climate parameter.

Main function	Study area	Technologies used	Results	Limitations	Citation
Short-term prediction of ETo	Arid and semi-arid regions of northern China	Propagation neural network model (BPNN)	The estimation results were very good for a period of up to seven days, but after that, the model's performance began to degrade.	The proposed model was unreliable at weather stations where minimum temperatures were lower than at other stations.	(Lu et al., 2023)
		Machine learning algorithms	The Catboost model outperformed other models in ETo predictions with the same input data.	Further study should consider evaluating model performance using ETo experimental data as targets.	(Y. Zhang et al., 2020)
	different climates in China	-KNN algorithm -Elman algorithm -Optimization algorithms	The study revealed that each model has the highest accuracy with the combination of surface radiation input (Rs) and temperature (T) input.	The proposed models did not respond in the same way in terms of accuracy in all the study regions.	(L. Zhao et al., 2022)
		-Extreme learning machine algorithm -Whale optimization algorithm (WOA)	The results showed that the optimized empirical Penman model presented the best estimate of ETo across China.	Other machine learning algorithms need to be tested to compare their performance with the one used in this research.	(Wu et al., 2022)
	different climates in Iran	Machine learning algorithms	The best accuracies were obtained especially in semi-arid climates.	Most models were wrong in estimating daily ETo in hyper-arid and humid climates, due to the great complexity of the evaporation process.	(Sharafi et al., 2023)
	different climates in Iran	- Machine learning algorithms -Optimization algorithms	The proposed hybrid model was able to estimate ETo values for the whole of Iran using the simplest climate variables, that is, temperature and wind speed.	The model's performance needs to be tested under conditions of climate change to improve results.	(Maroufpoor et al., 2020)
Short, medium and long-term prediction of ETo	different climates in China	-Extreme learning machine algorithm - Optimization algorithms	The proposed hybrid model made it possible to accurately estimate ETo on different time scales and with limited input data.	ETo's predictions for areas with climatic complexities were not good.	(Gong et al., 2021)
Long-term prediction of ETo	Coastal and inland areas in Italy	Machine learning algorithms	The established solution enabled a complete analysis and assessment of reference evapotranspiration trends.	The study was based on a small number of variables, which made it necessary to take other parameters into account.	(Di Nunno and Granata, 2023)

parameters and the best artificial intelligence model to use to accurately predict ETo.

RQ1.1: Does a good prediction of this coefficient guarantee good irrigation scheduling? In other words, is irrigation intensification always the optimal solution to compensate for the effect of soil evaporation, or do we need to think differently?

Some climatic changes such as the increase in atmospheric CO_2 concentration and temperature accelerate water loss through evaporation. The most obvious and simple solution to compensate for these losses is to increase irrigation intensity, but this is not always the most optimal solution. (Becker et al., 2023) conducted a research study in a heat-stressed region to see how the yield of certain crop types varied with increasing temperature. They found that the increase in CO_2 in the atmosphere of the study region had a positive effect on the yield of certain crop types, and so heat stress in this case led to a reduction in water demand. Nevertheless, these results could be different in the case of constantly rising temperatures. Another study carried out by (Karimzadeh Soureshjani et al., 2019) that addressed the existing relation between crop yield and temperature conditions came to the following result: management practices such as adjusting planting date can be applied by knowing the critical crop periods during which these crops are most vulnerable to climate change conditions.

According to these results, the irrigation scheduling is not only dependent on climatic changes due to weather conditions, but also on the biophysical responses of crops summarized in the crop coefficient Kc, which in turn is affected by several factors that are the physical, chemical, biological and hydrological characteristics of the soil.

RQ2: What factors have a significant influence on the crop coefficient Kc as mentionned in FAO Irrigation and Drainage Paper?

The value of the Kc coefficient varies considerably according to a number of factors including crop-specific factors such as crop type, season and growth stage, etc., soil-related factors such as the physical, chemical and biological properties of the soil, and agricultural practices. For precise and real-time irrigation management purposes, the Kc coefficient is driven by two coefficients, a base crop coefficient Kcb which describes the plant transpiration process and a soil water evaporation coefficient Ke (Anderson and French, 2019).

In the next part of this article, we'll try to shed light on the factors we believe have the greatest influence on determining the Kc coefficient. We will discuss and analyze new researches that use Internet of Things and Artificial Intelligence technologies for the detection and prediction of each factor.

RQ2.1: What effect do soil physical properties have on agricultural management in general? What new technologies are being used in this field of research?

Texture is a physical soil property quantified by the relative proportions of sand, clay and silt particles. It is these proportions that primarily indicate the physical quality of the soil in terms of hydraulic conductivity and water storage capacity, so determining soil texture is of paramount importance for planting and irrigation management purposes (Kurtulmuş et al., 2022).

(Inazumi et al., 2020) used a smartphone to capture images under different lighting conditions and soil surface appearances with the aim of identifying three soil types using a neural network model. The solution discussed allowed the identification of all the sand images, whereas the results were not good for the other soil types as they required adjustment of the learning parameters which was not easy for the authors to do given the limited performance of the equipment used. This study could be improved in this direction, and even for sandy soil detection by taking into account other conditions such as soil moisture.

The detection and classification of four clay soil types was the main focus of (J. and T., 2023) who captured images with a smartphone during all seasons and at different altitudes in every hectare of the study area to appropriately predict soil color characteristics. The proposed multi-stack ensemble model achieved very good results in terms of accuracy and precision.

(Al-Naji et al., 2021) took images of a set of silty soil samples at different times and light levels, via a video camera placed at different distances, to analyze it using a back propagation neural network for irrigation need prediction purposes. The proposed solution achieved ideal results. However, the system focused on a particular soil type and should be extended to other soil types. This was the aim of the study carried out by (Kurtulmuş et al., 2022) who relied on deep learning algorithms to distinguish between three classes of soil texture under different illumination and hydration conditions for irrigation scheduling. This work, considered to be the first work that used deep learning for texture-water class identification, can be improved by taking real-time image samples containing green plants with a wider range of conditions. Table 9 gives more details of the systems studied.

We can conclude that soil texture is an essential factor in precision agriculture. Researches aimed at identifying and classifying soil textures can be exploited in plantation management by choosing the type of crop that will give the best yield according to the physical properties of the field under study, as well as for irrigation planning and management purposes, since soil color can give an idea of the soil's hydric properties.

If we could come up with a model that encompasses all soil textures for crop management and irrigation purposes, then we could say that this model is the leader for this factor. Future works can be directed towards the creation of a model based on Artificial Intelligence and on Internet of Things technologies, that encompasses the totality of soil textures for crop management purposes and combines physical factors with soil chemical and biological factors for precise irrigation planning.

RQ2.2: What effect do soil chemical properties have on agricultural management in general? What new technologies are being used in this field of research?

Soil chemical properties are just as important as soil physical properties. Monitoring soil nutrient levels, cation exchange capacity and soil organic matter is of great importance as changes in these parameters directly influence crop coefficient and therefore yield (López-Arredondo et al., 2017). Fig. 6 shows the role of macronutrients, organic matter, cations and soil bacteria on plant development and growth.

Many studies have exploited new technologies to predict soil chemical properties. The research conducted by (Andrade et al., 2020) illustrates this. They used soil characteristic data acquired by portable X-ray fluorescence spectrometry (pXRF) to train machine learning algorithms to predict: total nitrogen, cation exchange capacity and soil organic matter. In general, the model results were acceptable. The samples taken concerned Eucalyptus plantations, so the model can be developed and improved by taking other: crop types, soil variabilities and environmental conditions.

(Srisomkiew et al., 2021) created digital soil maps to assess soil nutrient levels for the main reason of improving rice yield in the study area. The model developed in the study is MLR, which included soil data, terrain variables from a digital elevation model and satellite images collected under specific conditions to identify the most important spectral indices for prediction. Table 10 gives details of the systems studied.

Monitoring and predicting soil chemical properties also helps prevent major soil problems such as soil salinization. If soil salinity parameters exceed a certain level, plants will find it difficult to absorb water and nutrients and the evapotranspiration process may even stop. Predicting soil salinity to guide planting and irrigation was the main reason for the research conducted by (Xiao et al., 2023). They evaluated the performance of machine learning models to predict the ion composition of soil samples in an arid region of China. The results showed that the greater the number of input parameters the higher the accuracy of the model, but on the other hand, the lower the generalization capacity and stability. This study sees prospects in combining Internet of Things technologies with Artificial Intelligence algorithms for a more efficient and cheaper saline-soil monitoring.

Soil organic matter is a key factor determining variation in soil mechanical and hydrological properties (Hatanaka et al., 2022; Lin et al.,

Table 9
Studies using new technologies for soil texture detection and irrigation planning.

Main function	Target	Technologies used	Results	Limitations	Citation
Intelligent detection and classification of soil textures	Identify three types of soil: sand, clay and gravel	-Neural network model -Smartphone camera	The solution identified all the sand images without any problem.	The results were not good for the other soil types due to the limited performance of the computer used.	(Inazumi et al., 2020)
	Detection and classification of four types of clay soil	-Multiple-stack assembly model -Smartphone camera	The proposed model produced very good results in terms of accuracy and precision.	The model, according to the authors, can never be idealized due to noise in the input data or error sensibility in the model itself.	(J. and T., 2023)
Irrigation planning based on soil texture color	Predicting irrigation requirements for silty soils	-Video camera -Back Propagation Neural Network model	The proposed solution produced ideal results, as demonstrated by a coefficient of determination R^2=1.	The system has focused on one particular soil type and should be extended to other soil types.	(Al-Naji et al., 2021)
	Irrigation planning by distinguishing between three classes of soil texture: clay, silt and sandy silt	Deep learning algorithms	The proposed model led to very satisfactory results in terms of accuracy despite the fact that it was not the best in terms of detection speed.	This work can be improved by taking real-time image samples containing green plants with a wider range of conditions.	(Kurtulmuş et al., 2022)

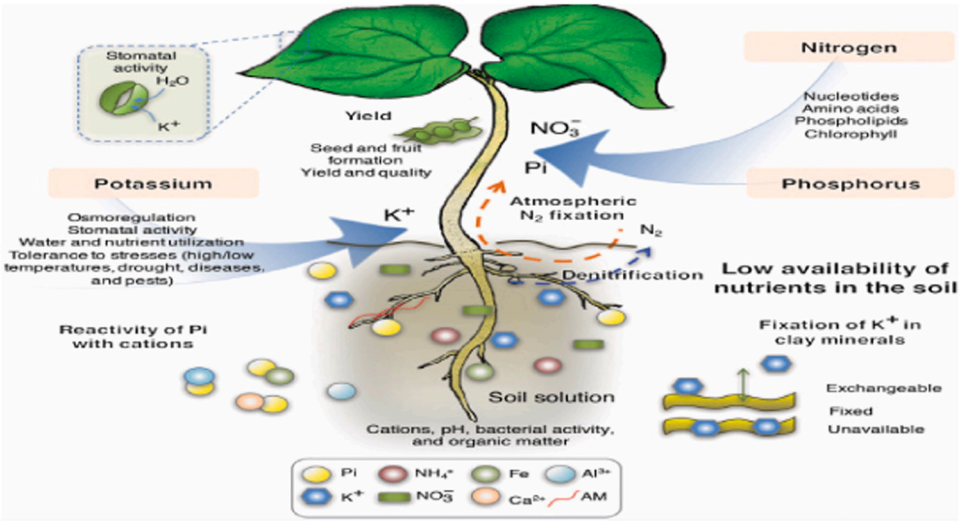


Fig. 6. The role of macronutrients in crop growth (López-Arredondo et al., 2017).

Table 10
Studies using new technologies to predict soil chemical properties.

Target	Technologies used	Results	Limitations	Citation
Prediction of total nitrogen, cation exchange capacity and soil organic matter	-Portable X-ray fluorescence spectrometry (pXRF) -Machine learning algorithms	In general, the model results were acceptable as this study is one of the few conducted on soil types under tropical conditions.	The proposed model was influenced by the highly dynamic behavior in the soil environment of the input variables caused by soil management.	(Andrade et al., 2020)
Assessment of soil nutrient levels	-MLR model -Digital elevation model -Satellite images	The digital maps generated made it possible to accurately assess various soil properties.	Some characteristics failed the test due to the timing of sampling which took place during a dry season.	(Srisomkiew et al., 2021)
Predicting the ion composition of soil samples to avoid soil salinity	Machine learning models	The model proposed with the input combination of soil temperature, soil water content and electrical conductivity was able to predict all soil salinity parameters with better accuracy, stability and generalizability than other models.	Model generalization and stability become poor with multiple input parameters.	(Xiao et al., 2023)

2022). (Lin et al., 2022) found that the improvement of soil mechanical properties (degree of compaction) is linked to the preservation or increase in soil organic matter content, while (Hatanaka et al., 2022) found that the alleviation of soil water deficit is linked to the accumulation of soil organic matter which contributes to moist, heavy metal-poor and nutrient-rich conditions.

We can conclude that the evapotranspiration process is also linked to the chemical properties of the soil. Maintaining a suitable level of these properties, according to agricultural needs, helps to increase yields by avoiding, on the one hand, problems linked to soil salinity and, on the other, mechanical and hydrological soil problems.

RQ2.3: What are the soil's hydrological properties to be taken into account when managing irrigation?

Accurate real-time irrigation scheduling requires the use of the dual crop coefficient approach Kc=Kcb+Ke, where Ke is the soil evaporation coefficient. Estimation of Ke requires daily monitoring and calculation

of soil water content in the upper arable layer (Anderson and French, 2019), hence the need to monitor soil moisture levels for irrigation management purposes.

RQ2.3.1: What new technologies are being used to estimate soil moisture?

(Chandrappa et al., 2023) used Internet of Things technologies presented by soil moisture sensors installed at different depths decided according to the depth of plant roots in the study site and installed at different zones in terms of water retention capacity of the study area in order to predict soil moisture. In addition to prediction, another aim of the study was to understand the relation between soil moisture and seasonal variation in meteorological data. The results showed a strong correlation between seasonally grouped input parameters and non-seasonal ones. Combining remote sensing and Internet of Things technologies to simulate soil moisture distribution was one of the authors' proposals for improving the study.

Integrating remote sensing data may also be helpful to overcome the problems of sensor data quality such as noise, errors and data discontinuity due to equipment failure or network problems (Abowarda et al., 2021).

The study conducted by (Togneri et al., 2022) could have achieved more accurate and precise results if it had included real-time remote sensing data on crop phenology such as leaf area index (LAI) and normalized difference vegetation index (NDVI), and not relied on traditional statistically defined coefficients to predict maximum and minimum next-days soil moisture values for reasons of monitoring the risk of extreme values. Table 11 gives details of the systems studied.

Remote sensing-based variables such as normalized difference vegetation index (NDVI) and vertical spatiotemporal soil moisture data from the Hydrus-1D physical model were used by (A et al., 2022) as input variables to the ConvLSTM model for continuous spatiotemporal prediction of root zone soil moisture with good large-scale accuracy.

Hourly, daily, weekly and even monthly soil moisture prediction was the objective of the study carried out by (Datta and Faroughi, 2022). They trained a multi-headed LSTM model with past soil moisture time series to estimate its future values. The proposed model was very efficient and can be improved by adding other physical attributes as input data.

In hydrology, estimating soil moisture is paramount as a key regulating element in understanding groundwater conditions (Masrur Ahmed et al., 2021). The relation between groundwater and soil moisture was discussed by (A et al., 2022), they showed that there is a strong correlation between groundwater level and soil moisture. The recharge and support functions between soil moisture and groundwater depend on the depth of the groundwater.

RQ2.3.2: What is the effect of groundwater on crop growth, and how can we exploit new technologies to estimate and predict groundwater levels?

We can't talk about agriculture without water resources, and among these resources, groundwater is of great importance in the agricultural sector. For example, the water balance of the root zone is modified by shallow groundwater, which has a more complex impact on crop growth and production (Chen et al., 2022).

(M. Han et al., 2015) found that groundwater level has a significant impact on crop growth in the study area as well as on various aspects of soil water balance. Cotton growth was influenced by groundwater levels at various locations, both favourably and unfavourably. Capillary rise in the water table is also strongly influenced by cotton growth. Consequently, groundwater is a vital element that needs to be taken into account in agricultural management. Groundwater level prediction is also used for irrigation planning purposes and to prevent overexploitation and depletion of underground water resources. Several studies have taken advantage of new technologies to predict groundwater levels for irrigation management purposes.

(Sheikh Khozani et al., 2022) have used a hybrid model that combines the ARIMA statistical model, the LSTM algorithm and optimization algorithms to capture the non-linear and linear components of a groundwater level time series for groundwater index prediction purposes over periods ranging from three to nine months without the use of climatic data. The proposed model is very useful in situations where modellers do not have access to meteorological data, but despite this valuable contribution, according to the authors, the model can be developed and improved either by integrating meteorological parameters to better understand the relation between these parameters and groundwater, or by relying on future scenarios for simulating meteorological parameters in the context of climate change and using other Artificial Intelligence algorithms coupled to the ARIMA model.

(Gharehbaghi et al., 2022) used, in addition to observed hydrological and groundwater depth data, regional meteorological data to predict groundwater levels in different arid and semi-arid agricultural zones in Iran. The final model proposed, entitled GRU2+, was deemed innovative and unique. Despite all this, the proposed model has certain limitations as it requires continuous measurements of the various inputs in addition to the difficulty of prediction, since groundwater level is a non-stationary, complex and inconsistent phenomenon.

The regional study conducted by (Cai et al., 2021) had two main objectives. The first objective was the prediction of groundwater levels using the GRU neural network, and the second objective was to study the impact of regional characteristics on improving prediction accuracy. The proposed model performed better in regions with climatic, geological and topographical features that favor more effective groundwater responses to dynamic external changes than in other regions where results were worse. This drawback could have been overcome if physical constraints had been included in the algorithms used to interpret the physical significance and hence generality of the model. The approach adopted by (Cai et al., 2022) addressed this need by

Table 11
Studies using new technologies for soil moisture prediction.

Target	Technologies used	Results	Limitations	Citation
Seasonal soil moisture prediction	-Soil moisture sensors -Machine learning and deep learning algorithms	The algorithms performed differently from season to season. Each season had a specific algorithm that performed better than the others.	Simulation of soil moisture distribution must be included in the study to determine the optimum location for sensors.	(Chandrappa et al., 2023)
Daily prediction of maximum and minimum soil moisture values	-Soil moisture sensors -Machine learning algorithms	The results showed that the LightGBM algorithm performed best in terms of accuracy and stability.	Maximum humidity is difficult to predict due to uncertainties linked to input parameters.	(Togneri et al., 2022)
Continuous spatio-temporal prediction of soil moisture in the root zone	-ConvLSTM model -Remote sensing data -Hydrus-1D physical model	The results of the model estimation, especially for deep layers, have increased significantly compared to the products of the GLDAS global system.	The study showed that there is a high correlation between groundwater level and soil moisture, but due to the lack of information, this level was not taken as a predictor variable in the ConvLSTM model.	(A et al., 2022)
Short- and long-term soil moisture prediction	Multi-head LSTM model	The model produced highly accurate results, with a coefficient of determination R^2=95.04% for future predictions.	The model can be enhanced by adding other physical attributes as input data.	(Datta and Faroughi, 2022)

integrating groundwater-related water balance equations as physical constraints in pure and hybrid deep learning architectures. Despite all the efforts made, the black-box nature of the proposed hybrid model is still not revealed due to the lack of detailed geographical information, insufficient hydrometeorological data and historical observations, and tedious manual adjustment of the learning parameters. The model therefore still needs to be improved to ensure its universality. Table 12 gives more details of the systems studied.

Groundwater quality for irrigation purposes is another area of study. In addition to the existence of groundwater, the quality of this water for irrigation purposes needs to be ascertained. (Liu et al., 2022) found that human activities affect the hydrochemical characteristics of groundwater. The increase in some chemical characteristics has reduced vegetation growth and limited plant cover, and worse than that, posed risks to human health. (Bennett, 2023) assessed the quality of groundwater used for irrigation in Tanzania. They found that the majority of groundwater in the study area is classified as poor to unsuitable for irrigation, so treatment is essential before use in irrigation to avoid low yields. The use of a tolerable rate of fertilizer in order to minimize the concentration of chemical components unsuitable for vegetation growth in the study area, is one of the solutions proposed.

RQ2.3.3: What is the effect of hydraulic conductivity on irrigation scheduling, and how can we exploit new technologies to estimate and predict hydraulic conductivity?

Saturated hydraulic conductivity and unsaturated hydraulique conductivity are other important parameters in irrigation scheduling. They are used to characterize many aspects of soil water flow such as infiltration, water balance, nutrient migration, irrigation and drainage system design and monitoring (D'Emilio et al., 2018).

Direct measurements cannot easily deduce these parameters under natural conditions, whatever the scale of interest (Rezaei et al., 2021) as they present a high degree of spatial heterogeneity (Fu et al., 2021). (Tian et al., 2018) have attempted, through the use of thermal pulse sensors and a simple water balance approach, to continuously estimate the fine depth-scale dynamics of unsaturated hydraulic conductivity under field conditions. Although this method has shown promising results, it has only been tested in a sandy loam soil and under simple meteorological conditions, hence the need for more in-depth studies under other field conditions, which is difficult due to the financial and practical constraints associated with routine hydraulic conductivity measurement.

Pedotransfer functions (PTFs) are indirect methods for estimating hydraulic conductivity from other easily measurable soil properties such as organic carbon, ph, hydraulic conductivity, etc. (Vereecken et al., 2010). (Mozaffari et al., 2022) made a comparison between visible-near-infrared spectroscopy with classical regression pedotransfer functions for the prediction of saturated and quasi-saturated hydraulic conductivity of calcareous soils. The results showed that the PTFs developed are more accurate in terms of hydraulic conductivity prediction. This better performance of PTFs can be linked to the use of different attributes such as soil structure properties and particle size distribution parameters.

The use of machine learning algorithms has become frequent in the development of TFPs. (Veloso et al., 2022) evaluated the performance of four machine learning algorithms in predicting saturated soil hydraulic conductivity and soil moisture PTFs at specific stresses in the Brazilian savannah. The results showed that regression trees were the best for predicting saturated hydraulic conductivity. (Albalasmeh et al., 2022) showed that the generalized regression neural network (GRNN) provided reliable predictions of saturated hydraulic conductivity in arid and semi-arid environments. (H. Kashani et al., 2020) developed and tested a hybrid model based on artificial neural networks to improve the accuracy of saturated soil hydraulic conductivity prediction in an arid region of Iran.(Aguilera et al., 2022) have exploited the advantages of artificial neural networks to develop a new methodology for mapping saturated hydraulic conductivity in the topsoil of their study area, based on quick and inexpensive auxiliary data.(Rezaei et al., 2023) have used remote sensing and field measured data for appropriate saturated hydraulic conductivity prediction and mapping in a sandy agricultural environment using machine learning algorithms based on trees. Table 13 gives more details of the systems studied.

The difficulty in predicting hydraulic conductivity is due to the fact that the soil properties that affect this parameter depend on the spatial scale (Fu et al., 2021). (W. Zhao et al., 2023) have shown that the predictive accuracy of saturated hydraulic conductivity PTFs is affected when the sample size is too small to cover the study area, leading to seriously biased estimates if the assumed relationship does not verify in the region of extrapolation, so they concluded that there is still a lack of saturated hydraulic conductivity prediction models universally applicable to different regions that have nearly the same characteristics.

RQ3: What is the effect of non-standard elements such as weeds and plant diseases on crop development and yield?

Another factor affecting the evapotranspiration process is the presence of pests and weeds. Functions essential to crop development such as photosynthesis and water and nutrient uptake are disrupted and limited in the presence of weeds and pathogens (Gao et al., 2021), sometimes leading to catastrophic yields.

Pest and weed detection is a highly advanced field that has benefited from the development of algorithms especially those of deep learning for the detection and classification of plant pests and diseases. Despite technological advances, this field still has certain shortcomings, which we will discuss below.

Table 12
Studies using new technologies to predict groundwater levels.

Target	Technologies used	Results	Limitations	Citation
Prediction of groundwater level index	-LSTM algorithm -ARIMA statistical model -Optimization algorithms	The model was able to provide accurate forecasts for current periods, while accuracy decreased with increasing forecast horizon.	The model can be expanded to include meteorological parameters to better understand the relation between these parameters and groundwater.	(Sheikh Khozani et al., 2022)
Groundwater level prediction in different arid and semi-arid agricultural areas in Iran	GRU-based neural network models	The model demonstrated its superiority over other state-of-the-art algorithms in terms of accuracy, simplicity and efficiency of long-term groundwater level prediction.	The proposed model has certain limitations as it requires continuous measurements of the various inputs, in addition to the difficulty of prediction due to the fact that groundwater levels are a non-stationary phenomenon.	(Gharehbaghi et al., 2022)
Groundwater level prediction and study of the impact of regional characteristics on improving prediction accuracy	GRU neural network	The proposed model performed better in regions with characteristics that favor more effective groundwater responses to dynamic external changes.	The model performance depended on the characteristics of the study area.	(Cai et al., 2021)
	Pure and hybrid deep learning architectures	The results showed the superiority of the proposed hybrid model over pure deep learning models, making the model more accurate and interpretable.	The black-box nature of the proposed hybrid model has yet to be revealed.	(Cai et al., 2022)

Table 13
Studies using new technologies to predict saturated hydraulic conductivity.

Target	Technologies used	Results	Limitations	Citation
Assessing the performance of machine learning algorithms in predicting saturated soil hydraulic conductivity Ks	-Machine learning algorithms	Algorithms based on regression trees obtained better results for Ks prediction. In addition, the predictor set with soil texture, structure and moisture variables showed the best results compared to the other predictor sets.	The models overestimate low Ks values, while the property is underestimated at higher Ks values. This can be attributed to the high spatial variability of Ks and the relatively small number of samples used in this study.	(Veloso et al., 2022)
Evaluate the potential use of a generalized regression neural network to identify the optimal set of soil characteristics for predicting Ks in arid and semi-arid environments.	Generalized regression neural network (GRNN)	Soil texture parameters and electrical conductivity improved Ks prediction capability. In addition, the GRNN model provided reliable predictions of Ks.	The predictive accuracy of Ks in sandy soils was not good due to the small sample size of this particular class.	(Albalasmeh et al., 2022)
Develop and test a hybrid intelligent model based on a multiple model strategy (MM-ANN) for soil Ks estimation.	-Machine learning algorithms	The results showed that the proposed model serves as an effective pedotransfer function for predicting soil Ks using readily available soil parameters as input variables.	There were relatively high model estimation errors in the western part of the study area.	(H. Kashani et al., 2020)
Develop a new methodology for mapping topsoil Ks based on rapid and inexpensive auxiliary data.	Artificial neural networks (ANN)	The proposed model is robust and outperforms all the reference PTFs tested, and the methodology presented provides an inexpensive representation of the Ks distribution.	The under-performance of the model is due to the anthropization of soils, hence the need to further improve the model by including other terrain attributes and spectral data.	(Aguilera et al., 2022)
Exploit remote sensing data with limited soil data and tree-based machine learning models to predict soil Ks.	-Tree-based machine learning algorithms. - remote sensing	The results of the RF model are robust and can therefore be applied to Ks mapping.	The model led to slight overestimates and underestimates of Ks.	(Rezaei et al., 2023)

RQ3.1: What new technologies are being used to manage weeds?

Uncontrolled weeds can adversely affect the soil evaporation process. The use of herbicides is the solution used by farmers to control the spread of weeds. This solution can be a double-edged sword if not adopted rationally. Excessive use of herbicides can increase weed resistance to them and may even affect soil characteristics and food security. The study carried out by (Dang et al., 2023) focused on this aspect of precise weed location for rational herbicide application. They attempted to compare the performance of seven versions of YOLO object detectors for multi-class detection of weeds in the Cotton crop in real time. The robustness of the models can be improved by including more representative training data and optimized training strategies, as well as

Table 14
Studies using new technologies to manage weeds and plant diseases.

Main function	Sub area of study	Technologies used	Crop type and cultivation environment	Results	Citation
Weed management	Real-time weed location	-7 versions of YOLO object detectors -Digital and Smartphones cameras	Cotton arable fields.	The models offered differed in terms of precision, speed and complexity but in general, all versions offered great performance.	(Dang et al., 2023)
	Weed location and detection in predictable and unpredictable scenarios	-YOLOv3 detection model -Industrial cameras	Arable fields of sugar beet and potatoes.	The proposed model considerably reduced the generalization error, but the change in the type of camera used and the noise caused by fallen tree leaves affected the model's performance.	(Ruigrok et al., 2023)
	Weed seed detection	-6 CNN models -Industrial image acquisition system	140 weed species collected from the wild and processed in an industrial environment.	The performance of the models differed in terms of accuracy, cost and speed, and the choice of the best model depended on the specific requirements of use. In general, the best detection accuracies were linked to seeds with clear shape characteristics.	(Luo et al., 2023)
Plant disease management	Real-time detection of plant diseases using a mobile application	-Simplified model of MobileNetV2 architecture -Digital and Smartphones cameras	-Images of apple tree leaves from an orchard. -Images of tomato, citrus and apple leaves from data available on the internet.	The proposed lightweight deep learning model showed high detection accuracy with less computational resource requirements. Since the proposed model requires only a small amount of storage space, several models can be integrated into the same application.	(Janarthan et al., 2022)
	Detection and classification of diseased leaves according to disease level and severity.	-Image segmentation algorithms -DL model based on Kohonen network and INC-VGGN architecture	Images of various crop types acquired from the Plant Village and Plant Doc databases.	The results of the proposed model were better in terms of sensitivity, specificity and accuracy.	(Pal and Kumar, 2023)
	Detecting the type and the degree of a plant disease	-Transformer based models -RGB camera	-Real-time images of cucumber plants and banana leaves. -Images of potato and tomato disease leaves from controlled environments.	The results showed that the proposed model is robust, able to determine with high accuracy the type of diseases and their degree in complex contexts containing several noise sources and in stable contexts.	(Guo et al., 2022)
	Classification of the type and severity of plant diseases simultaneously and separately	Model based on a three-branch Swim transformer classification network	31,718 disease images acquired from the AI Challenger 2018 dataset.	The proposed model performs very well in disease classification with an accuracy of 99%, while for severity classification, the accuracy is 88.73%.	(Yang et al., 2023)

by testing other YOLO models. Table 14 gives more details of the systems studied.

Developing a general location and detection model that can be used in all predictable and unpredictable field scenarios is another hot topic. (Ruigrok et al., 2023) have tried, through the use of the YOLOv3 detection model, to study the effect of increasing the number of training images as well as their quality on the generalization of the weed detection model in order to decrease the error caused by generalization. They found that the addition of new image samples covering new variations led to an improvement in generalization, whereas the simple addition of images brought no added value in terms of generalization.

Detection between the seeds of several weed species and crop seeds is the objective of the study conducted by (Luo et al., 2023). They proposed a system for detecting and classifying 140 weed species using sensors and deep learning algorithms trained on image samples containing several phenotypic characteristics of the seeds. Expanding the dataset is a proposal for improving this study.

RQ3.2: What new technologies are being used to manage plant diseases?

Accurate and rapid identification of plant diseases is a necessity in the agricultural sector. (Janarthan et al., 2022) have proposed a mobile application for the detection of diseases in different types of crops, the mobile application proposed does not require internet connectivity to operate. The generality of the application can be ensured by integrating active learning and incremental learning to classify unknown diseases, this is the first perspective of the study, the second perspective is to ensure the interpretation of results and recommend solutions to farmers.

Taking measures according to the severity of diseased plants is a solution that limits the excessive use of pesticides. (Pal and Kumar, 2023) have proposed a model that ensures the detection and classification of diseased leaves according to disease severity level. Future work in this area will focus on: detecting multiple diseases in an image, knowing in addition to disease severity the stage of the disease in order to take suitable actions and using other samples from real-life scenarios.

(Guo et al., 2022) used data acquired under natural and controlled environmental conditions to detect the type and the degree of a disease. Complexity and large size are the two drawbacks of the model, and these drawbacks increase both: the memory and the hardware requirements of the proposed model. Improvements need to be made in this area.

(Yang et al., 2023) have proposed a model that enables the classification of disease type and severity simultaneously and separately. The model can be improved and generalized by taking a larger dataset and using it on mobile applications.

So, we can conclude that pest and weed management is of great importance. The spread of weeds and pathogens simultaneously affects both evaporation and transpiration processes, leading to low yields.

RQ4: Do the farming practices adopted in a field affect the crop coefficient, and if so, how does this manifest itself?

The appearance of weeds can be caused by agricultural practices that are sometimes not adapted to the agro-system. (Souza et al., 2020) studied the effect of two irrigation systems on weed establishment and crop yield in the study area. They found that the mode of application and the quantity of water, directly affect weed dynamics, and these disturbances affect yield and the effectiveness of irrigation practices. It is therefore necessary to choose the appropriate irrigation method (micro-sprinkler, drip, etc.), according to the crop's water requirements, and to ensure that it is placed in an optimal location (Sawadogo et al., 2023).

Crop water requirements also differ according to the type of crop and more specifically, its growth stage. Crops generally pass through four growth stages: initial stage, crop development stage, mid-season stage and end-of-season stage (Anderson and French, 2019). Each growth stage has its own irrigation requirements, which is why Kc crop coefficient values change from one stage to the next. Precise estimation of the Kc coefficient for each crop type according to these growth stages, for irrigation planning purposes, remains a difficult operation up to now since several factors, as discussed earlier, enter into and affect the crop evapotranspiration process. Few studies have attempted to estimate the crop coefficient Kc using new technologies.

(Shao et al., 2021) used the leaf area index (LAI) and six multi-spectral vegetation indices extracted from images acquired by an unmanned aerial vehicle (UAV) to train them in machine learning algorithms for field-scale estimation of the Kc value of the maize crop. The proposed model had a coefficient of determination R^2=0.65. The study by (Shao et al., 2023) is an improvement of the previous study, they added four other multi-spectral vegetation indices and also took water stress into consideration. The results showed a coefficient of determination R^2=0.65. Further improvement is therefore required by including other factors and other crop types.

A good understanding of the characteristics of agricultural zones enables farmers to adapt the most suitable cultivation practices to their needs, resulting in excellent yields while optimizing water use.(W. Zhang et al., 2022) analyzed and evaluated the effect of combining mulching and drip irrigation on crop yields. The use of mulch in the study area produced good results in terms of crop yield and water use efficiency compared with areas where mulch was not included. The choice of the type of mulch to be applied depends on the climatic, crop and hydrological characteristics of the study area.

5. Discussion

The dual crop coefficient method is widely used for irrigation scheduling (Anderson and French, 2019). Estimating and forecasting the parameters involved in the calculation of this method has so far proved difficult despite all the technological advances we are currently experiencing (Shao et al., 2021, 2023).

The first question of our research was about the climate parameter ETo, its importance and the new technologies used for the estimation and prediction of this coefficient. We concluded that the models proposed in this direction are not global in character, and therefore more extensive studies need to be carried out. Models need to be developed for each region of the world, indicating the best input parameters and the best artificial intelligence model to use to accurately predict ETo.

These results led us to another research question which discusses the effect of predicting this parameter on irrigation scheduling while shedding light on the solution of irrigation intensification to compensate for the effect of climatic parameters on the soil and its feasibility in all cases. We found that irrigation intensification is not always the optimal solution, sometimes other agricultural practices such as adjusting planting date by knowing the critical crop periods during which these crops are most vulnerable to climate change conditions are more suitable and give more efficient results than irrigation intensification (Karimzadeh Soureshjani et al., 2019). At this point we came to the conclusion that it is important to study, in addition to meteorological effects, the biophysical responses of crops for good irrigation management.

The biophysical responses of crops are summarized in the crop coefficient Kc which in turn changes according to several factors leading to the second research question which discusses the factors that influence the Kc coefficient. Soil physical, chemical and hydrological properties are assumed to be the key factors directly affecting crop growth (Anderson and French, 2019).

The first subquestion in this path was about soil texture and its role in agricultural management in general, and about the existence of studies using new technologies in this field of research. We found that soil texture plays a vital role in planting and irrigation management (Kurtulmuş et al., 2022) and that researches in this field using new technologies are still in their infancy. The proposal of an intelligent model that encompasses the totality of soil textures for planting and irrigation management purposes will undoubtedly be of great benefit to the agricultural sector.

The second subquestion was about soil chemical properties, their importance and the new technologies used in this field of study. We

found that monitoring soil nutrient levels and organic matter is essential for both selecting crops that match the chemical properties of the field and managing the quantities of nutrients to be applied during the crop development phase (Srisomkiew et al., 2021) as well as combating serious problems that negatively affect the evapotranspiration process and yield and contaminate the soil (Xiao et al., 2023). Models that combine connected object technologies and artificial intelligence algorithms to manage the type and timing of planting according to field properties and to monitor soil nutrient levels in real time according to various crop types remain to be improved.

The third subquestion was about the hydrological properties of the soil that need to be taken into account when managing irrigation. We discussed three hydrological properties: the water-holding capacity of the soil referred to in the article as soil moisture, the soil permeability which led us into a discussion of groundwater levels and quality and finally the hydraulic conductivity. So this question generated three others.

The first was about new technologies for estimating soil moisture. We found that new technologies are widely used in this field of research, but elaborate studies should be carried out to plan irrigation needs for each growing season according to all the factors mentioned above.

The second was about groundwater, its impact on crop growth and the new technologies used to estimate and predict groundwater levels and quality. We found that groundwater directly feeds the roots of certain plants if it is not deep (Chen et al., 2022; M. Han et al., 2015) and that farmers rely heavily on this water for irrigation, hence the need to monitor its level for rational planning of water resources over the short, medium and long term. The domain of groundwater level monitoring using new technologies presents several difficulties due to the non-stationarity and complexity of this phenomenon (Gharehbaghi et al., 2022). An accurate model that gives correct interpretations in this domain will be very useful and beneficial, but this can only be guaranteed with detailed and precise informations about the study environment (Cai et al., 2022).

Groundwater quality is another area of research. Excessive use of fertilizers, pesticides and herbicides is adversely affecting the quality of these waters. In this case, the use of these resources presents serious dangers to human health and prevents the proper growth of crops, hence the need to check the quality of these waters before use (Bennett, 2023).

The third was about hydraulic conductivity and the new technologies used to estimate saturated and unsaturated hydraulic conductivity.The conclusion found by (W. Zhao et al., 2023) is that there is still a lack of hydraulic conductivity prediction models universally applicable to different regions with the same characteristics because this parameter presents a high degree of spatial heterogeneity (Fu et al., 2021).

The rational use of fertilizers, pesticides and herbicides can guarantee good water quality for irrigation. This point led us to the third research question which in turn led us to two other subquestions all of which discuss the effect of non-standard elements such as pests, weeds and plant diseases on crop development and yield, and the new technologies used to detect, locate and classify these elements with the aim of limiting the use of pesticides and herbicides. We have found that deep learning algorithms are being used extensively in this field of study for real-time monitoring of weeds, pathogens and degrees of plant diseases and their severities, but despite all these advances, there are still avenues of research to be explored such as real-time learning of field circumstances (Janarthan et al., 2022), interpretation of results and proposal of recommendations for problems found according to disease stage, degree and severity (Pal and Kumar, 2023).

Sometimes inappropriate farming practices can stimulate the appearance of plant pests and diseases, hence the final research question which studies the effect of farming practices on the crop coefficient. We have found that the right choice of farming practices can achieve the desired yield while optimizing the use of natural resources (Souza et al., 2020).

The profitability of a crop field is closely linked to the right strategies adopted in a specific crop plot. The beneficial and favorable effect of water on crop income has led us to propose the following approach, modeled in Fig. 7, which aims to plan and manage irrigation efficiently.

Choosing the right irrigation strategy is a key link in the agricultural production chain. Our model aims to use new Internet of Things and Artificial Intelligence technologies to understand and manage the various characteristics of the field, i.e. meteorological, physical, chemical and hydrological characteristics in order to choose the appropriate type of crop in the field of study and then schedule irrigation according to the type of crop chosen and the field characteristics studied. The agricultural production process does not stop at this point, as control and monitoring of non-standard conditions and disruptive elements, as well as the choice of suitable cultivation practices, are necessary to achieve the best possible yield from the field under study.

Fig. 7 shows the different approaches adopted for irrigation and farm management in our systematic literature review. For each property studied, we have mentioned the data used to arrive at the desired property prediction (green colored part in model), the sources from which the data were acquired e.g. IoT technologies, weather stations etc, (orange colored part), the artificial intelligence algorithms used to predict the output variables (blue colored part), the variable to be predicted (gray colored part) and finally the future research directions (red colored part).

6. Conclusions

In this study, we systematically reviewed the scientific literature of the last eight years to explore the new Internet of Things and artificial intelligence technologies used, on the one hand, to monitor and predict the climatic, physical, chemical and hydrological characteristics of a field and, on the other, to detect, locate and prevent the biological characteristics of a field. To our knowledge, this is the first SLR of its kind, and the first to examine the use of new technologies to predict and monitor every field characteristic in order to manage irrigation efficiently. Adopting an SLR as a method of answering our main research questions has proved to be a very useful choice, and has enabled us to obtain significant information that could be useful to both the committee of researchers and the stakeholders. We have identified that the profitability of a crop field is closely linked to the right strategies adopted in a specific crop plot, and these strategies can only be defined after a good understanding of the field's characteristics. For the climatic properties of the field, the ETo parameter was studied. For the physical properties of the field, the soil texture classes and their irrigation requirements were discussed. For the chemical properties of the soil, the nutrient level and soil chemical properties were discussed. For the hydrological properties of the soil, soil moisture, hydraulic conductivity and groundwater level and quality were studied. For the biological properties, the detection and localization of weeds and plant diseases were studied. This sytematic literature review has enabled us to develop an intelligent model that brings together the different approaches adopted for irrigation scheduling and to pave the way for further research.

This systematic literature review focuses on the field scale, it can be expanded by covering essential irrigation management issues for large-scale irrigation areas or large-scale farmland. Finally, we hope that the contributions of this review will help anyone interested in the field of irrigation management to form a holistic and clear view.

CRediT authorship contribution statement

Afaf Bounajra: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Kamal El Guemmat:** Visualization, Validation, Supervision, Methodology, Conceptualization. **Khalifa Mansouri:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Fatiha Akef:** Validation, Supervision, Investigation.

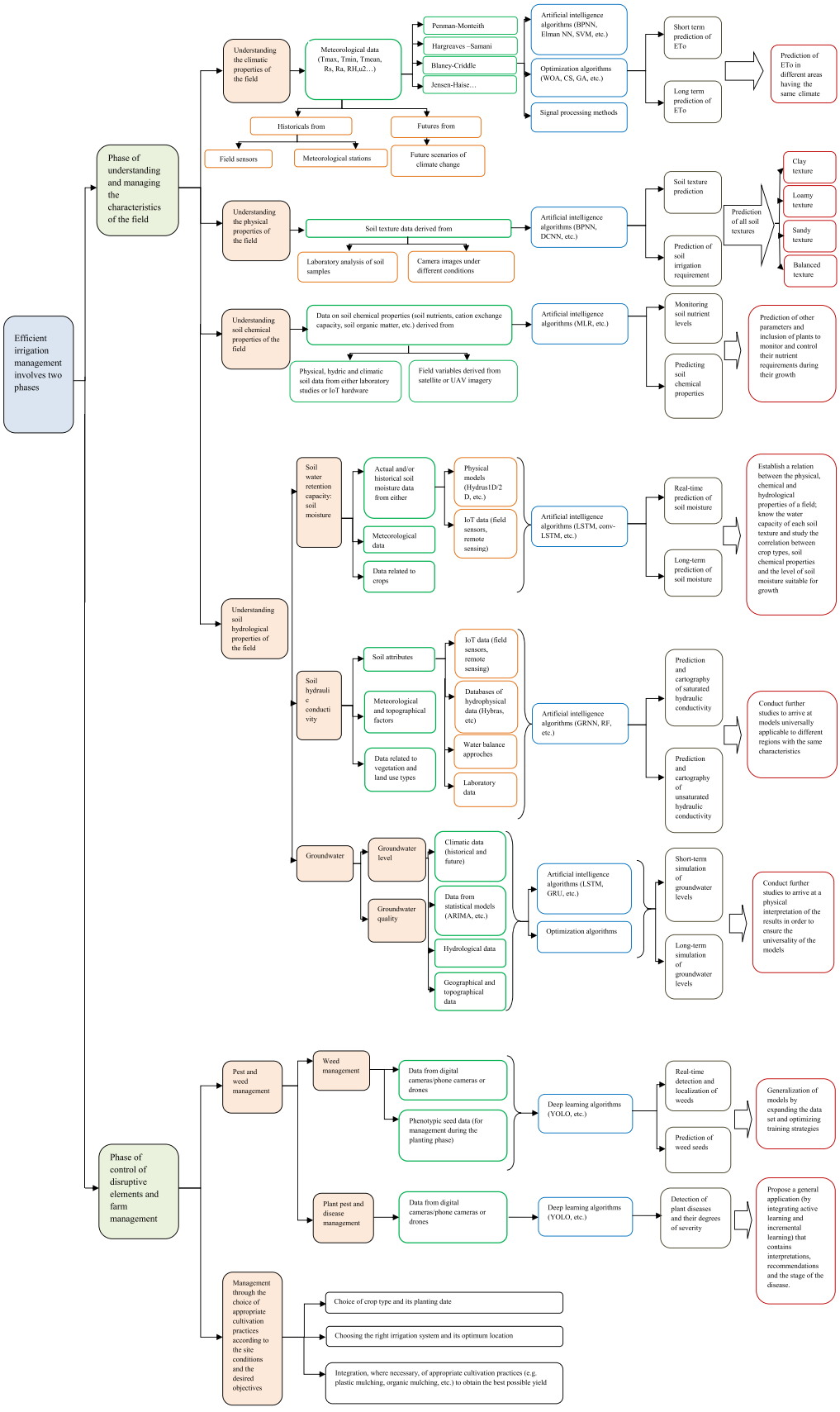


Fig. 7. model of different approaches of irrigation through the use of new technologies.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

Appendix A. Data extraction table

Table 6
The data extraction table used in this SLR.

	Extraction element	Contents
General information		
1	Title	Full article title
2	Year	Year of publication
3	Authors	
4	Repository	Science Direct, Springer, IEEE Xplore, Scopus,Web of Science
Study description		
5	Targeted domain	
6	Study motivation	
7	Main aims of the study	
8	Study area	
9	Data used	
10	Range of data	
11	Material used to acquire data	
12	Data processing methods	
13	Artificial intelligence algorithms used	
14	Findings	
15	Limitations	
16	Future Works	
Evaluation		
17	Personal note	Reviewer opinion about the study
18	Additional note	Publication details
19	Quality assessment	Quality scores

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